



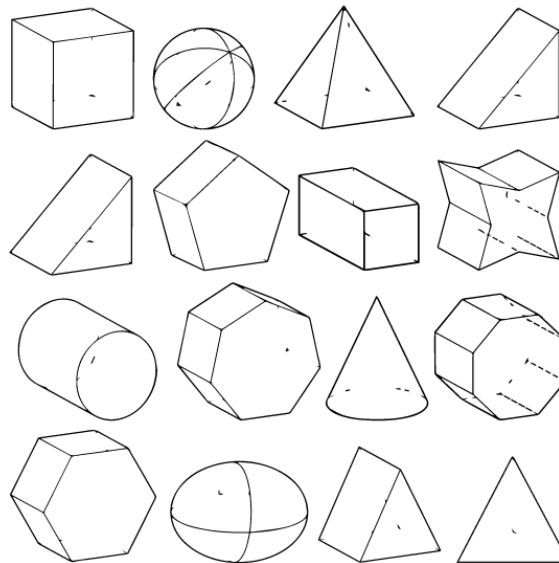
Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points

**Let's Start!****Shapes in Programming****Learning Outcome:**

Exchange information about relational database management system.

Ask your partner:

- What do you know about data models?
- Have you used ER diagrams before? How?

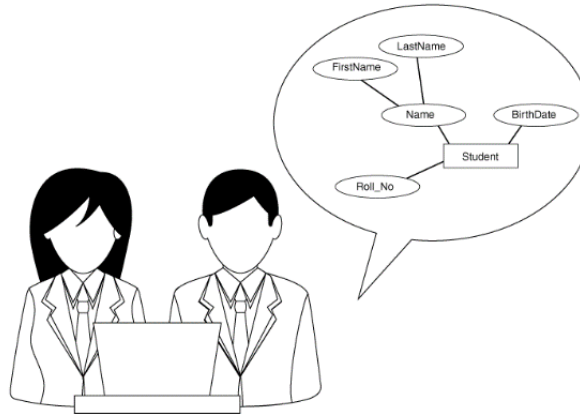
Can Do Statements:

- I can identify the shapes to use in diagrams with entities and attributes.
- I can talk about how to depict: composite, multivalued, and derived attributes.
- I can discuss binary relationships and cardinality through the use of diagrams.



Let's Explore!

Mary explains to Andrew how to understand a diagram.



Instructions: Read the text below.

Andrew: Hi, Mary. How are you?

Mary: Good, Andrew. What about you? You don't look as good as yesterday. Stop drinking coffee! It's not as good as you think.

Andrew: It's not coffee. I'm analyzing a diagram, but I don't understand it. Can you help me?

Mary: Let me see.

Andrew: Hmm. Why are all the shapes different?

Mary: This is very simple. Pay attention; all the entities are rectangles, all the attributes of the entities are ellipses, and all the relationships have the shape of a diamond.

Andrew: Oh, now I see. You are right! But, what about these dashed lines and double line ellipses. What do they mean?

Mary: Right! The dashed line ellipse is a derived attribute and the double ellipse ones are multivalued attributes.

Andrew: I get it now, but I have a few more questions: why are those ellipses connected to other ellipses?

Mary: It is because some attributes have other attributes as well; it is called a composite attribute. It is like your full name; it has a first and a last name, which are the composite attributes.

Andrew: I still have one question. It is very hard for me to describe the relationships on these diagrams. Can you explain them to me?

Mary: Look! The relationship is represented by the diamond-shaped box, and all the entities that are part of that relation are connected to it by lines. A binary relationship happens when two entities are involved.

Andrew: And what about the cardinalities?

Mary: Those are instances that are part of a relationship. For example, it might be that only one instance of each entity is associated in the relation. It can also be one-to-many, many-to-one, or many-to-many.

Andrew: Thanks for your help!

**Let's Practice!**

Mary explains to Andrew how to use an ER diagram.

Instructions: Read the paragraph. Underline all the mistakes in the paragraph.

Used rectangles to represent entities. Names them with the entity they represent. Takes into account that you use ellipses to represent attributes. Connected the attribute with a simple line to its entity. Remembering that if the attribute is composite, you need to represent every node as an ellipse, and then connect it to its attribute.

**Let's Practice!**

ER Diagram Representation is as not as difficult as you may think.

Instructions: Circle the correct option to complete the sentences.

1. Relationships are (as easy to be represented / as easy represented) as entities; use a diamond-shaped box.

2. You can connect (as much as / as many attributes as) you want by using lines.

3. Creating diagrams is (not as difficult / as difficult as) as you may think.

4. An attribute is as (as not as important / important as an) entity since it represents the unique characteristic of an entity.

5. A multivalued attribute can have (as many as / as many values) as necessary, depending on the number of characteristics.



Extension Activity

Using an ER Diagram Representation

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Think of people that are close to you (family, classmates).
- Create entities based on them.
- Create an ER Diagram.
- Add attributes to each entity.
- Think on any composite and derive attributes as possible.
- Make a poster with your ER diagram.
- Give an oral presentation to explain your diagram.

Self-Assessment

Yes

No

- I create entities based on people I know.
- I add attributes to each entity.
- I add composite and derive attributes.
- I make a poster with my ER diagram representation.
- I give an oral presentation to explain my diagram.

☐	☐
☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- dashed line** (noun): a line made up of dashes
e.g. The dashed line ellipse is a derived attribute and the double ellipse ones are multivalued attributes.
- connected** (adjective): something linked together subsequently
e.g. Why are those ellipses connected to other ellipses?
- involved** (adjective): referring to a situation, implicated
e.g. A binary relationship happens when two entities are involved.
- entity** (noun): an object or a graphical representation of data
e.g. Pay attention; all the entities are rectangles.
- ellipse** (noun): an oval-shaped curve
e.g. All the attributes of the entities are ellipses.
- attribute** (noun): a particular feature of an entity in a diagram
e.g. All the attributes of the entities are ellipses



Learning Points

Imperatives

The **imperative form** is used to give advice, instructions, commands, or orders.

An imperative sentence begins with the base form of the verb. A negative imperative begins with **don't** and the base form of the verb.

An imperative sentence ends with a period (.) or an exclamation mark (!).

Example:

Base verb form	Complement
Connect	entities with their attributes with lines.
Create	associations between the entities.
Don't forget	that weak entities depend on others.

Examples

- *Pay attention; all the entities are rectangles, all the attributes of the entities are ellipses, and all the relationships have the shape of a diamond.*
- *Look! The relationship is represented by the diamond-shaped box, and all the entities that are part of that relation are connected to it by lines.*

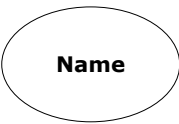
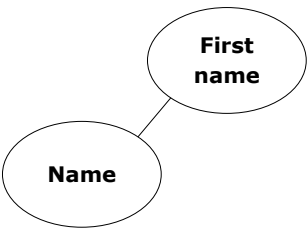
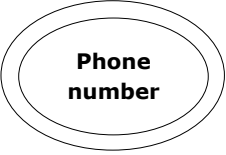
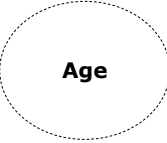


Learning Points

Shapes in Diagrams

When creating an ER diagram, there are specific shapes used for specific purposes.

Let's look at the chart to see the use of each shape:

Entities are represented by rectangles.	
Attributes are represented by ellipses.	
Composite attributes are represented by ellipses connected to other ellipses.	
Multivalued attributes are represented by double ellipses.	
Derived attributes are represented with a dashed-line ellipse.	
Relationships are represented by a diamond.	